# IGCSE 0607 Practice Questions — E2.1

## Introduction to algebra

**Total Questions:** 12 — **Total Marks:** 59 Marks

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**1** Find the value of $3a - 2b$ when $a = 4$ and $b = -5$.

..... **[2]**

**2** Given the formula $v = u + at$.

Find the value of $v$ when $u = 5$, $a = -2$, and $t = 3$.

$v =$ ..... **[3]**

**3** **(a)** Work out the value of $5x - 2y$ when $x = 3$ and $y = -4$.

..... **[2]**

**(b)** Evaluate

$$\frac{a^2 - b^3}{2c}$$

when $a = -4$, $b = 2$ and $c = 3$.

..... **[2]**

**4 (a) (a)** The cost of hiring a plumber is calculated using the formula

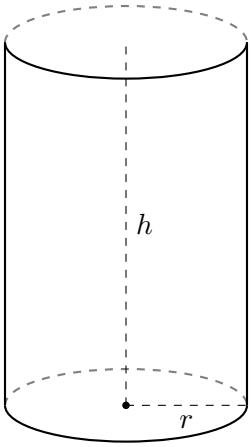
$C = 40 + 25h$ where $C$ is the cost in dollars and $h$ is the number of hours worked.
Work out the cost of hiring a plumber for 3.5 hours.

$C = \$ \ldots\ldots\ldots\ldots\ldots$ **[2]**

**(b)** Given that $v = u + at$, find the value of $v$ when $u = -12$, $a = 9.8$ and $t = 3$.

$v = \ldots\ldots\ldots\ldots\ldots$ **[3]**

**5** A cylindrical container of radius $r$ and height $h$ is open at the top.
The total exterior surface area of the container, $A$, consists of the curved surface area and the circular base.




**(a)** Show that the total surface area $A$ can be written as

$$A = \pi r(2h + r)$$

**[2]**

**(b)** **(i)** Calculate the exact surface area $A$ when $r = 4$ and $h = 6$.
Give your answer in terms of $\pi$.

$A =$ ..... **[2]**

**(ii)** The volume of the cylinder is $V$.
The formula for the volume is $V = \pi r^2 h$.
Express $h$ in terms of $V$, $\pi$ and $r$, and substitute this into the expression for $A$ to show that

$$A = \frac{2V}{r} + \pi r^2$$

**[3]**

**6** Consider the expression $P = \dfrac{2x^2 - 3y}{x - y}$.

**(a)** Calculate the value of $P$ when $x = -2$ and $y = 4$.

$P =$ ..... **[3]**

**(b)** Rearrange the formula to make $y$ the subject.

$y =$ ..... **[4]**

**7** The cost, $C$ dollars, of hiring a plumber for $h$ hours is given by the formula:

$C = 45 + 35h$

**(a)** Calculate the cost of hiring the plumber for 3.5 hours.

$C =$ ..... **[2]**

**(b)** The plumber charges a total of \$325 for a job.
Calculate the number of hours the plumber worked.

$h =$ ..... hours **[3]**

**8** The volume, $V$, of a spherical cap of height $h$ and base radius $a$ is given by the formula

$$V = \frac{1}{6}\pi h(3a^2 + h^2)$$

**(a)** Calculate the volume of a spherical cap with height $h = 4.2$ cm and base radius $a = 6.5$ cm. Give your answer correct to 1 decimal place.

$V =$ ..... cm$^3$ **[2]**

**(b)** Rearrange the formula to make $a^2$ the subject.

$a^2 =$ ..... **[3]**

**(c)** Find the base radius, $a$, of a spherical cap with volume $V = 150$ cm$^3$ and height $h = 3$ cm. Give your answer correct to 3 significant figures.

$a =$ ..... cm **[1]**

**9** A formula used in physics is

$$s = ut + \frac{1}{2}at^2$$

**(a)** **(a)** Calculate the value of $s$ when $u = 5.2$, $t = 4.5$ and $a = -9.8$.

$s =$ ..... **[2]**

**(b)** Calculate the positive value of $t$ when $s = 40$, $u = 15$ and $a = -9.8$.

$t =$ ..... **[3]**

**10** **(a)** Work out the value of $4x - 3y$ when $x = -2$ and $y = 5$.

..... **[2]**

**(b)** Work out the value of $\dfrac{p^2 - 7}{2q}$ when $p = 5$ and $q = -3$.

..... **[2]**

**11** The volume, $V$, of a cone with radius $r$ and height $h$ is given by the formula

$$V = \frac{1}{3}\pi r^2 h$$

**(a)** Calculate the value of $V$ when $r = 6.4$ and $h = 12.6$.
Give your answer correct to 1 decimal place.

$V =$ ..... **[2]**

**(b)** Work out the value of $r$ when $V = 800$ and $h = 14$.
Give your answer correct to 3 significant figures.

$r =$ ..... **[3]**

**12** The distance, $d$, between two points $(x_1, y_1)$ and $(x_2, y_2)$ is given by the formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

**(a)** Work out the value of $d$ when $x_1 = 4$, $y_1 = -3$, $x_2 = -8$, and $y_2 = 2$.

$d =$ ..... **[2]**

**(b)** Given that $d = 17$, $x_1 = k$, $y_1 = 8$, $x_2 = 3$, and $y_2 = -7$, work out the two possible values of $k$.

$k =$ ..... , $k =$ ..... **[4]**